Cut Sheet

Board Design Summary

Style: tumbling-block-3d
Dimensions: 12.000" (12") × 18.000" (18") × 1.500" (1 1/2")
Species: Walnut, Maple

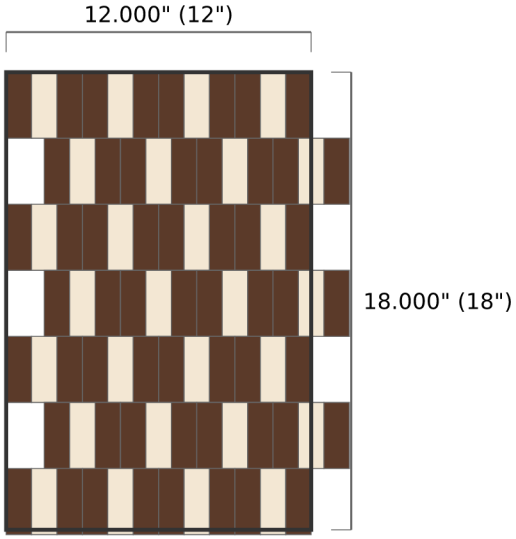
Cut List

Part	Width	Thickness	Length	Species
Board strips	1.000" (1")	1.500" (1 1/2")	18.000" (18")	Walnut

Cut List

Part	Qty	Width	Thickness	Length	Species
Panel A, sub-panel 1/2: Walnut/Maple strip-set panel: glue 6 whole strips (order: W M W M W M), rip each strip 1-1/16" (27 mm), then joint to 1" (25.4 mm) wide; Planer OK: 6-1/4" (158.4 mm) wide, long grain; then crosscut sled with stop block (never fence) 7 slices × 1-11/16" (43 mm) each (sled must clear 6-1/4" (158.4 mm)); Too wide for the jointer (5-31/32" (152 mm)): flatten one face on a planer sled first, then plane the other face parallel	1	6-1/4" (158.4 mm)	1-27/32" (46.6 mm)	14-23/32" (374 mm)	Walnut/Maple
Panel A, sub-panel 2/2: Walnut/Maple strip-set panel: glue 6 whole strips (order: W M W M W M), rip each strip 1-1/16" (27 mm), then joint to 1" (25.4 mm) wide; Planer OK: 6-1/4" (158.4 mm) wide, long grain; then crosscut sled with stop block (never fence) 7 slices × 1-11/16" (43 mm) each (sled must clear 6-1/4" (158.4 mm)); Too wide for the jointer (5-31/32" (152 mm)): flatten one face on a planer sled first, then plane the other face parallel	1	6-1/4" (158.4 mm)	1-27/32" (46.6 mm)	14-23/32" (374 mm)	Walnut/Maple

Board Layout (Top View)



Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

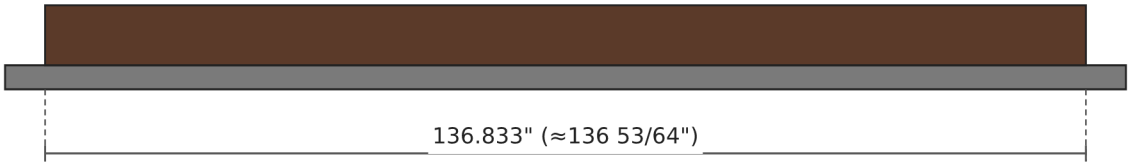
Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.7 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

3D Tumbling Block (End Grain): step 1: Mill rough stock

Whole blank on the jointer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 271.697" (≈271 45/64") long, 56 slices of 4.692" (≈4 11/16") crosscut after milling.

Maple strip: blank 136.833" (≈136 53/64") long, 28 slices of 4.692" (≈4 11/16") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 2: Rip strips

Rip three species to equal width with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

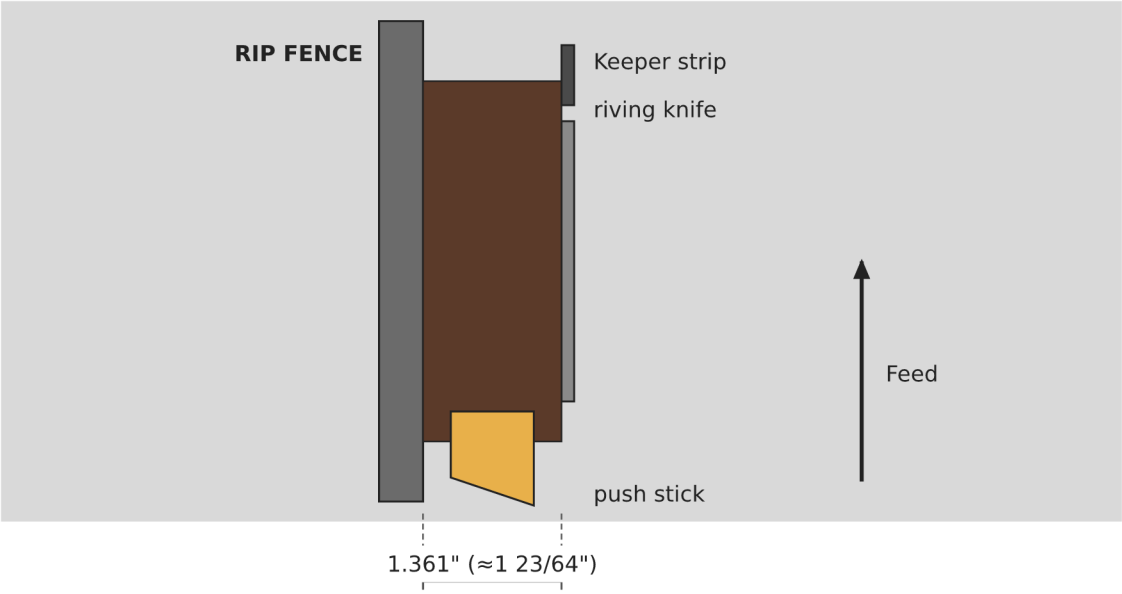
Settings:

Fence to blade: 1-1/16" (27 mm)

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

3D Tumbling Block (End Grain): rip-fence setup, step 2: Rip strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade

Rip three species to equal width with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

Step 3: Bevel strips

Rip a 30° bevel on both edges to make 60° diamonds in section. Use a hold-down; the offcut side must be the outside of the blade.

Settings:

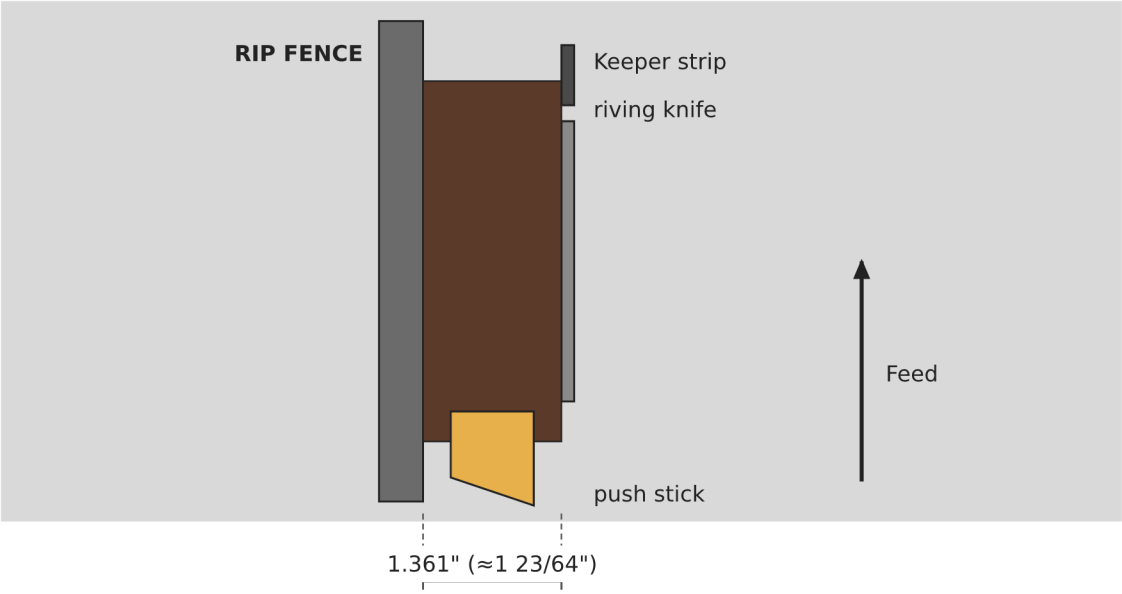
Blade tilt: 30°

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

WARNING: Make angled cuts on a sled or miter gauge with a hold-down, never against the rip fence alone.

3D Tumbling Block (End Grain): rip-fence setup, step 3: Bevel strips



Measure fence face → near side of a tooth

set toward the fence

Kerf: 1/8" (3.2 mm) full-kerf blade

Rip a 30° bevel on both edges to make 60° diamonds in section. Use a hold-down; the offcut side must be the outside of the blade.

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WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

WARNING: Make angled cuts on a sled or miter gauge with a hold-down, never against the rip fence alone.

Step 4: Glue up

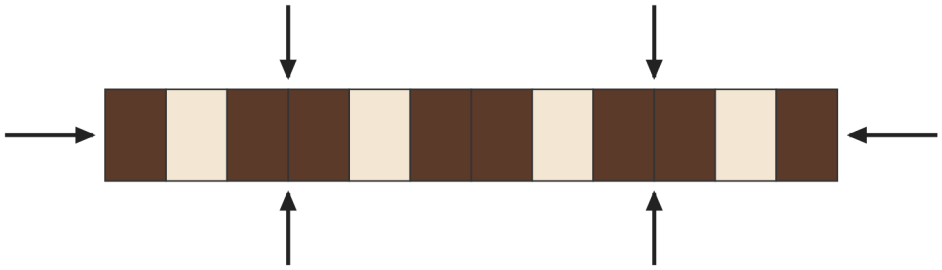
Glue the diamonds into hexagon logs, one of each tone. Alternate clamps above and below the panel and check for flat with a straightedge.
Flatten Panel A, sub-panel 1/2, 6-1/4" (158.4 mm) wide:
Planer OK: 6-1/4" (158.4 mm) wide, long grain.
Flatten Panel A, sub-panel 2/2, 6-1/4" (158.4 mm) wide:
Planer OK: 6-1/4" (158.4 mm) wide, long grain.

Settings:

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

3D Tumbling Block (End Grain): glue-up 1, step 4: Glue up



Glue the diamonds into hexagon logs, one of each tone. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

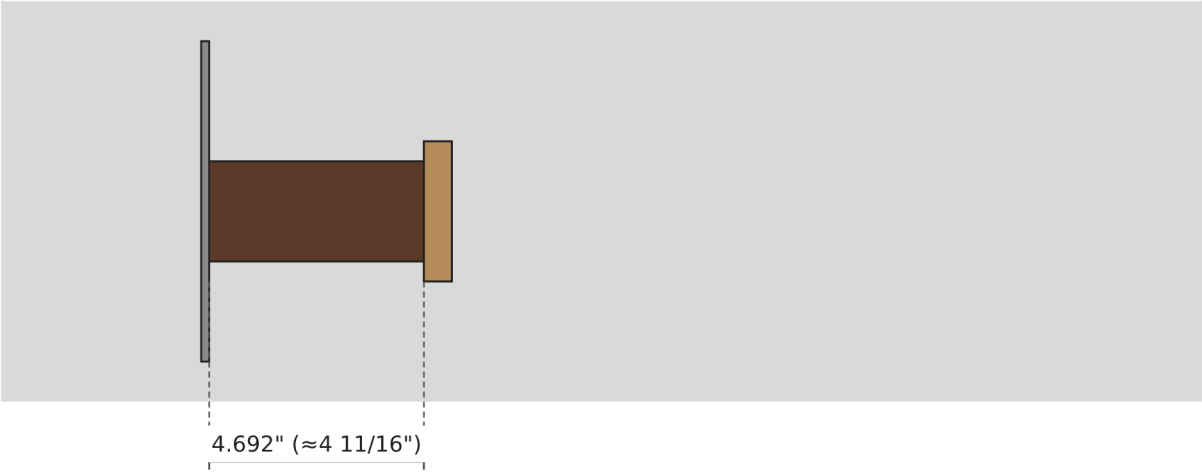
Step 5: Crosscut slices

Crosscut the logs into slices on a crosscut sled with a stop block.
Slice size: 1-11/16" (43 mm).
Final thickness: 1-1/2" (38 mm).
End-grain allowance: 5 mm.
Panel A, sub-panel 1/2 (order: W M W M W M): sled must clear 6-1/4" (158.4 mm).
Panel A, sub-panel 2/2 (order: W M W M W M): sled must clear 6-1/4" (158.4 mm).

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

3D Tumbling Block (End Grain): crosscut, step 5: Crosscut slices



Crosscut the logs into slices on a crosscut sled with a stop block.

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Step 6: Glue up

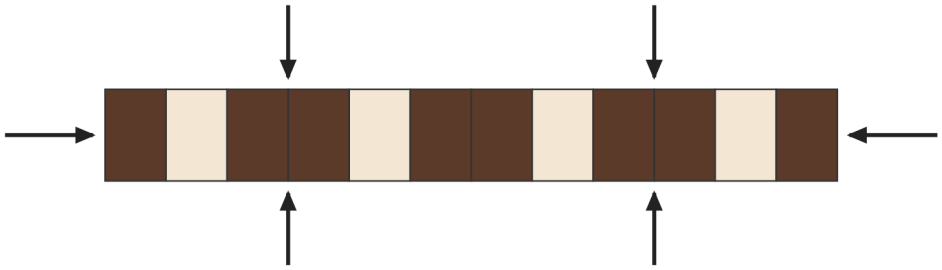
Glue hexagon slices into the final panel, then trim the zig-zag edges square. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

3D Tumbling Block (End Grain): glue-up 2, step 6: Glue up



Glue hexagon slices into the final panel, then trim the zig-zag edges square. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Step 7: Flatten and sand

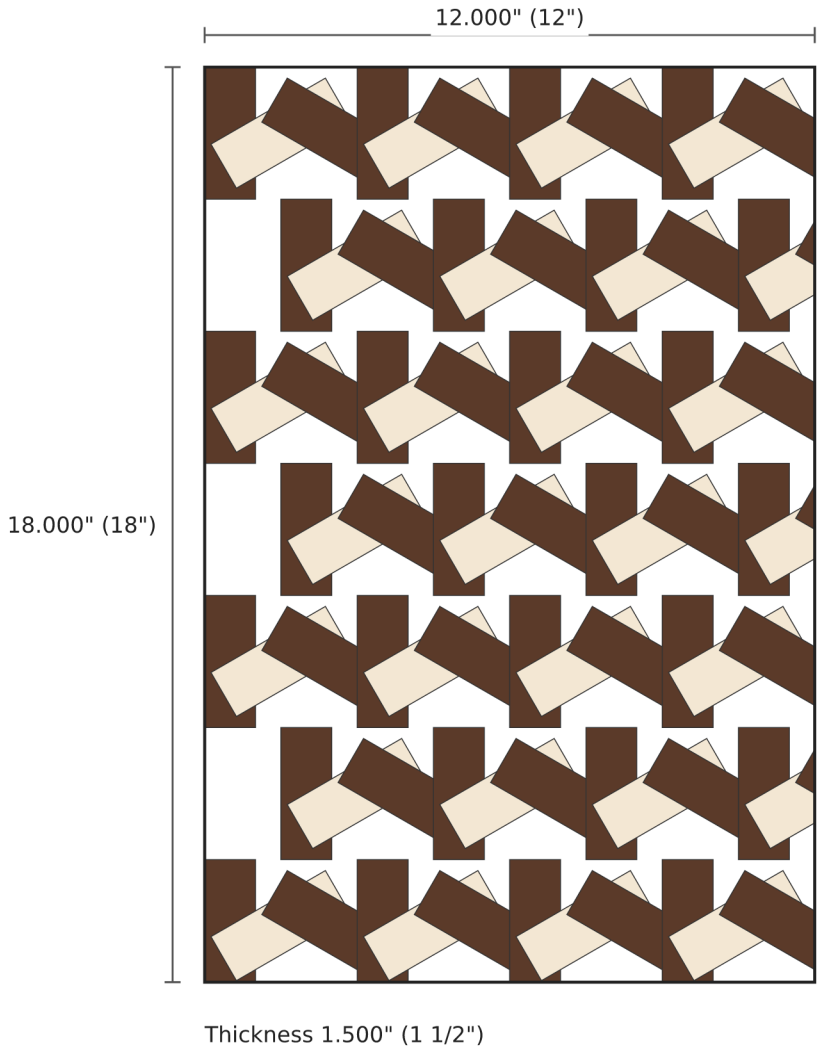
Flatten the final glue-up, 12" (304.8 mm) wide:
mixed grain: router sled or drum sander, never the planer.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

3D Tumbling Block (End Grain): top view



Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Step 8: Edge profile

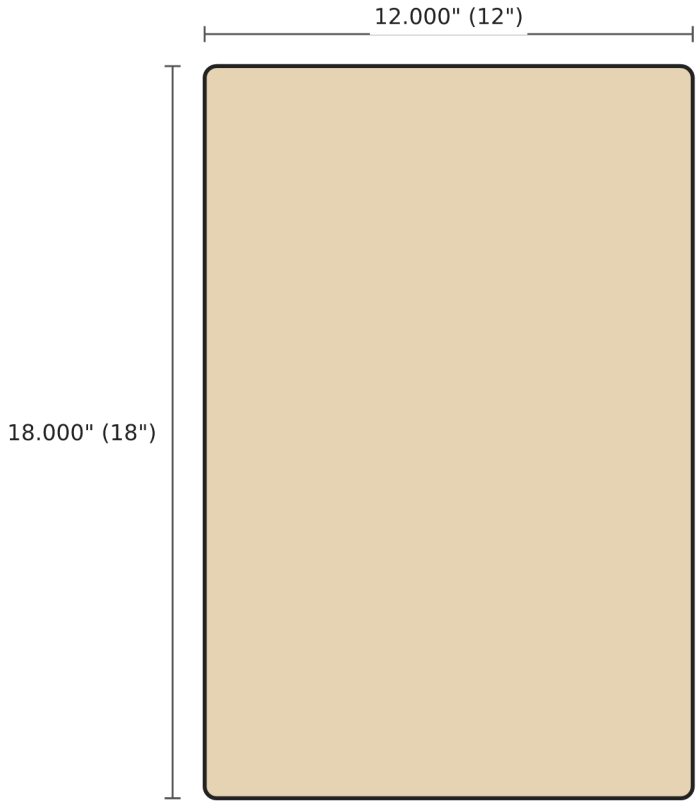
Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

Settings:

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

3D Tumbling Block (End Grain): outline, step 8: Edge profile



Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 9: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.